GTM965500P

Test Report

Model Level Model

GlobTek Engineering

Generated: April 16, 2026

Test Summary

Project: GTM965500P | Model Level Model | Total Test Items: 11

Status	OK / Resolved	For Review	Regression Req'd	In Progress	Has Issue	Invalid/ Incomplete Test	Unknown
Count	1	3	0	0	1	0	6

Model Level

Test Item	Status	Link
Input		
Input Current	Marked for review	View →
Inrush Current	Marked for review	View →
Power Factor	Marked for review	View →
Main Output		
Power Good (PG)	OK / Resolved	View →

Tests Not Yet Run

7 tests have not yet been run.

Model Level

Test Item	Status	Link
Main Output		
Remote-Off (RO)	Unknown	View →
Standby Output		
Output Voltage Regulation	Unknown	View →
Fan Output		
Output Voltage Regulation	Unknown	View →
Protections		
Under-Voltage Protection (UVP)	Unknown	View →
Over-Temperature Protection (OTP)	Unknown	View →
EMC		
Lightning Surge	Unknown	View →
PFC		
PFC Voltage Drop at 90Vac - overload	Has Issue	View →

Detailed Test Results

Model Level

Model Level	Input Current	OPEN	Marked for review

Specification / Procedure

Spec: 6.5A max. @ 100Vac, 5.2A typ. @ 115Vac, 2.5A typ. @ 240Vac (full load)

Suggest using the 12V unit since it has the lowest efficiency and will have the highest input current.

Results

Vision314 · 2026-04-16 [LATEST]

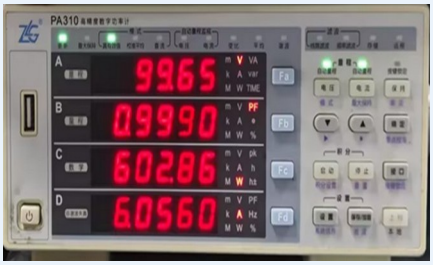
GTSZ

12V

Spec (A)	Reading (A)	Result	Picture
6.5	6.128	PASS	

24V

Spec (A)	Reading (A)	Result	Picture

6.5	6.056	PASS	

Mode I Level	Inrush Current	OPEN	Marked for review

Specification / Procedure

Spec: < 35A @ 115Vac , < 70A @ 240Vac (cold or warm start)

Use Chroma AC source, set start phase angle to 90°. Verify and record AC turn-on waveform, showing 90° turn-on with power supply detached. Connect power supply, Using oscilloscope current probe, measure and record peak value of inrush current.

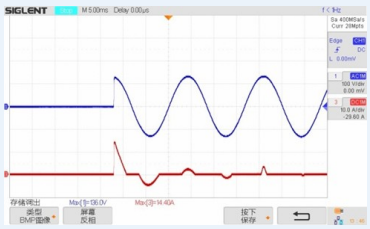
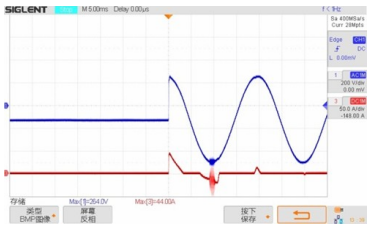
Inrush current should not depend on output voltage so any voltage can be used.

Results

Vision314 · 2026-04-16 [LATEST]

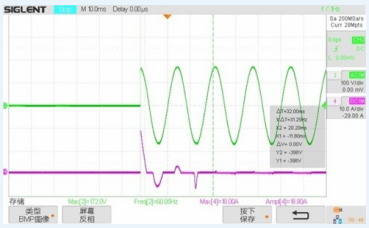
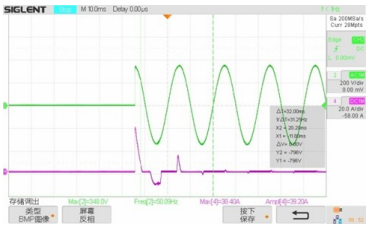
[GTSZ]

12V

Voltage	Spec (A)	Reading (A)	Result	Picture
115Vac	35	14.4A	PASS	
240Vac	70	44A	PASS	

24V

Voltage	Spec (A)	Reading (A)	Result	Picture

115Vac	35	18A	PASS	
240Vac	70	38.4A	PASS	

Mode I Level	Power Factor	OPEN	Marked for review

Specification / Procedure

Spec: > 0.98 @ 115Vac, > 0.97 @ 240Vac typ. (25 - 100% load)

Using Yokogawa WT3000, measure power factor at 25%, 50%, 75% and 100% load, at 115Vac and 240Vac.

Power factor does not depend on output voltage so any voltage can be used.

Results

Vision314 · 2026-04-16 [LATEST]

GTSZ

24V, 115Vac, Average Efficiency = 92.102%

Parameter	100%	75%	50%	25%
Percent of Rated Load	100%	75%	50%	25%
True Power Factor (Watts/VA)	0.9992	0.9990	0.9987	0.9975
Efficiency	91.763%	92.566%	92.886%	91.193%

24V, 230Vac, Average Efficiency = 94.247%

Parameter	100%	75%	50%	25%
Percent of Rated Load	100%	75%	50%	25%
True Power Factor (Watts/VA)	0.9987	0.9967	0.9920	0.9819
Efficiency	94.652%	94.625%	94.415%	93.295%

12V, 115Vac, Average Efficiency = 91.907%

Parameter	100%	75%	50%	25%
Percent of Rated Load	100%	75%	50%	25%
True Power Factor (Watts/VA)	0.9985	0.9989	0.9988	0.9974
Efficiency	90.751%	91.875%	92.853%	92.151%

12V, 230Vac, Average Efficiency = 93.553%

Parameter	100%	75%	50%	25%
Percent of Rated Load	100%	75%	50%	25%
True Power Factor (Watts/VA)	0.9985	0.9960	0.9907	0.9760
Efficiency	93.438%	93.786%	94.010%	92.976%

Mode I Level	Power Good (PG)	CLOSED	OK / Resolved

Specification / Procedure

Spec: Signal typically goes low ~10ms ahead of AC power failure or entering of protection state
5V logic level, VH = 4.90V typ. @ IPG= 100µA

Select any model output voltage. PG behavior should not depend output voltage.

Select a nominal input voltage (e.g. 115Vac)

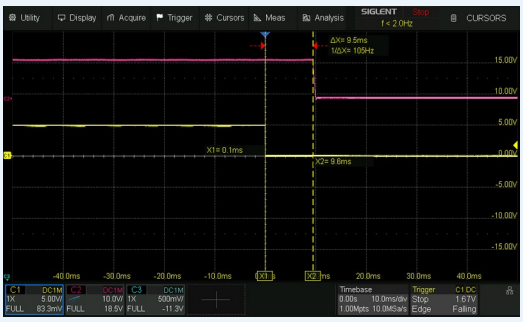
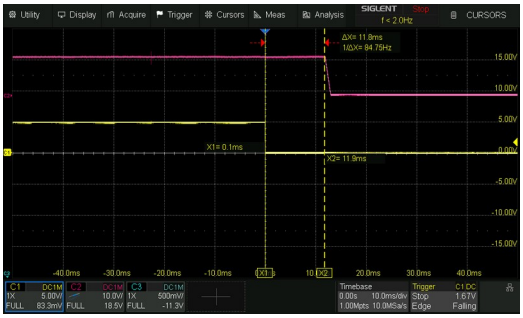
At OPP: Step the load above the OPP threshold. Measure the time between PG asserting low and the output voltage going low.

At AC power failure: At 75% load, switch off AC input power. Measure the time between PG asserting low and the output voltage going low.

Results

Vision314 · 2026-04-14 [LATEST]
YELLOW - PG Signal
PINK - Output Voltage

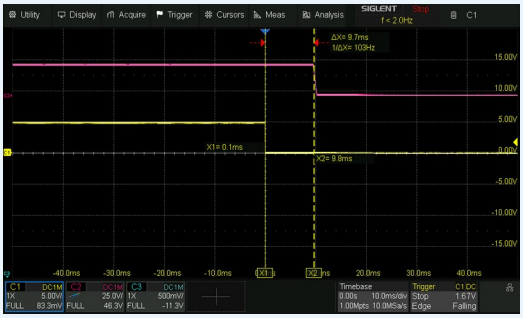
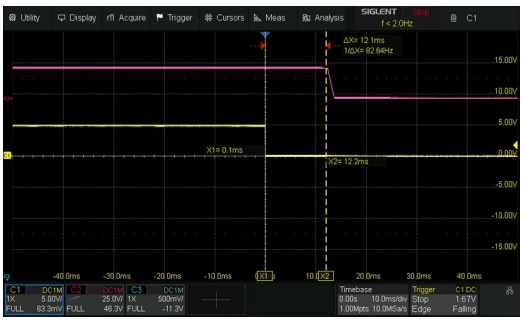
12V

Failure Mode	PICTURE	PG Time
OPP		9.5ms
AC Power Failure		11.8ms

OPP Load = 66A (144%)

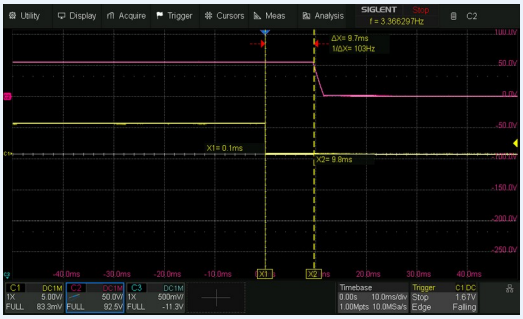
75% load = 34.4 A

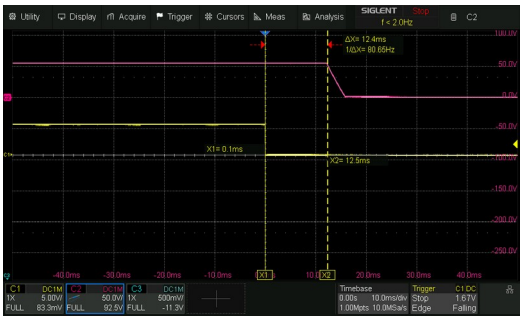
24V

Failure Mode	PICTURE	PG Time
OPP		9.7ms
AC Power Failure		12.2ms

OPP Load = 32A (139%)
75% load = 17.2 A

54V

Failure Mode	PICTURE	PG Time
OPP		9.7ms

AC Power Failure		12.4ms

OPP Load = 13A (128%)
75% load = 7.6 A

Untested — Specifications

Model Level

Model Level	Remote-Off (RO)	OPEN	Unknown

Specification / Procedure

Spec: Pull 'Remote Off' to COM to disable main output; leave floating or connect to +5VSB otherwise, IRO = 2mA typ. during disabled/off state

Select a nominal input voltage (e.g. 115Vac).

Connect signal breakout board with RO switch. Capture RO signal going low at switch off. Capture V_out going low in response.

Model Level	Output Voltage Regulation	OPEN	Unknown

Specification / Procedure

Spec: ± 5.0% (set point tolerance + load/line/cross regulation)

At nominal line voltage (e.g. 115Vac), set main-output load to 50% load, measure the standby output voltage:

1. Fan output = 0% / Standby output: 0%
2. Fan output = 0% / Standby output: 100%
1. Fan output = 100% / Standby output: 0%
2. Fan output = 100% / Standby output: 100%

Then, disable main output via Remote Off (RO) switch, change input voltage to 90Vac, and record the standby output voltage:

1. Fan output = 0% / Standby output: 0%
2. Fan output = 0% / Standby output: 100%
1. Fan output = 100% / Standby output: 0%
2. Fan output = 100% / Standby output: 100%

With Remote Off (RO) still enabled (main output off), change input voltage to 264Vac, and record the standby output voltage:

1. Fan output = 0% / Standby output: 0%
2. Fan output = 0% / Standby output: 100%
1. Fan output = 100% / Standby output: 0%
2. Fan output = 100% / Standby output: 100%

Model Level	Output Voltage Regulation	OPEN	Unknown

Specification / Procedure

At nominal line voltage (e.g. 115Vac), set main-output load to 50% load, measure the fan output voltage:

Fan output = 0% / Standby output: 0%
Fan output = 0% / Standby output: 100%
Fan output = 100% / Standby output: 0%
Fan output = 100% / Standby output: 100%

Then, disable main output via Remote Off (RO) switch, change input voltage to 90Vac, and record the fan output voltage:

Fan output = 0% / Standby output: 0%
Fan output = 0% / Standby output: 100%
Fan output = 100% / Standby output: 0%
Fan output = 100% / Standby output: 100%

With Remote Off (RO) still enabled (main output off), change input voltage to 264Vac, and record the fan output voltage:

Fan output = 0% / Standby output: 0%
Fan output = 0% / Standby output: 100%
Fan output = 100% / Standby output: 0%
Fan output = 100% / Standby output: 100%

Model Level	Under-Voltage Protection (UVP)	OPEN	Unknown

Specification / Procedure

Spec: Off @ 70Vac typ., Restore @ 80Vac typ.

Select any voltage model.

At light load (e.g. 10%), slowly reduce the input voltage until the unit shuts off. Record the voltage at which it turns off. Raise the input voltage slow and record the voltage at which it turns back on.

At full load, slowly reduce the input voltage until the unit shuts off. Record the voltage at which it turns off. Raise the input voltage slow and record the voltage at which it turns back on.

Check that there is no chatter and that the transitions are clean.

Model Level	Over-Temperature Protection (OTP)	OPEN	Unknown

Model Level	Lightning Surge	OPEN	Unknown

Model Level	PFC Voltage Drop at 90Vac - overload	OPEN	Has Issue

Specification / Procedure

Issue: At 90Vac, the PFC fails to maintain 395Vdc output at/near full load and above. Currently, it is understood that the SNSMAINS signal is faulty/noisy, causing TEA2376DT to erroneously reduce voltage loop gain, causing the PFC output voltage to fall out of regulation. There are protections indicated. The PFC "does not think anything is wrong", but the ship is sinking.

Mentioned here: https://github.com/GlobTek-Engineering/GTM965500P/issues/66#issuecomment-4164434018

Below: Output voltage (yellow), PG signal (blue), PFC voltage (magenta). Conditions: Approaching 145% load @ 90Vac input

[image: https://github.com/user-attachments/assets/74615967-e722-4f3c-8a99-d9e7e29c6439]

Improvement was gained by adding an RC filter to SNSMAINS which allowed operation at 100% load / 90Vac, but the issue was still not totally eradicated because the phenomenon again arises at overloads exceeding roughly 145%. Perhaps this is sufficiently beyond the specified range to safely ignore, but on the other hand, it's persistence still suggests a sensitivity issue that may rear its head in other ways later.